

Can-AI-Societies-Reach-Stability-A- Stochastic-Study-of-LLM-Agents:

A Stochastic Dynamical Framework for Memory,
Randomness, Adaptation, and Emergent Behavior in
LLM Agents

Jatin Aggarwal

2/07/2026

Abstract

Large language models (LLMs) are increasingly deployed as autonomous agents that interact repeatedly with other agents in shared environments. As these systems scale from small agent teams to large synthetic societies, collective behavior cannot be understood solely through the properties of individual agents. Cooperation, coordination, polarization, behavioral persistence, and systemic instability may emerge from interactions among memory, stochasticity, adaptation, population size, and interaction structure.

This paper proposes a mathematical and computational framework for identifying the *stability frontier* of LLM-based artificial societies. We model each agent as a stochastic, history-dependent decision process whose behavior is governed by five principal system variables: population size N , memory horizon M , stochasticity T , interaction frequency F , and adaptation rate α . Rather than assuming rational best-response behavior, agents select actions according to conditional policies that depend on their current state and finite interaction history.

We introduce four primary population-level observables: cooperation C , behavioral stability S , adaptation velocity A_{vel} , and strategic diversity D . These are supplemented by measures of volatility, autocorrelation, perturbation resilience, recovery time, and polarization. We propose a controlled experimental program spanning repeated social dilemmas and coordination games under multiple interaction topologies.

The central theoretical object is an empirically estimated stability surface,

$$\Phi(M, T, N, \alpha, F, \mathcal{G}, \mathcal{H}) = 0,$$

where $\mathcal{G}$ denotes the interaction topology and $\mathcal{H}$ denotes the game environment. The framework does not assume that a universal critical threshold exists. Instead, it provides a methodology for testing whether reproducible dynamical phase boundaries emerge, whether they depend

on game and network structure, and whether they generalize across LLM families and prompting regimes.

The proposed framework connects behavioral game theory, evolutionary dynamics, complex systems, and generative-agent simulation. Its primary contribution is a shift from evaluating isolated agent behavior toward measuring the dynamical stability, resilience, and phase structure of entire LLM populations.

Keywords: large language models; multi-agent systems; generative agents; artificial societies; game theory; cooperation; dynamical systems; complex systems; phase transitions; behavioral stability; emergence; agent-based modeling

1 Introduction

Large language models have evolved from systems primarily evaluated through static question answering into components of autonomous systems capable of planning, tool use, memory, communication, and repeated interaction. Recent generative-agent architectures demonstrate that language models can be embedded in environments containing persistent agents, observations, memory, reflection, and social interaction [9]. Large-scale simulation platforms have subsequently demonstrated the feasibility of constructing synthetic populations containing thousands of LLM-driven agents and millions of interactions [10].

These developments create a fundamental scientific problem.

When a single LLM agent makes a decision, its behavior can be evaluated through conventional benchmarks. However, when hundreds or thousands of agents interact repeatedly, the relevant object of study is no longer the individual model. It is the *collective dynamical system* generated by the population.

An artificial society can exhibit properties that are not obvious from the behavior of any individual member. For example, individually cooperative agents can collectively defect if negative experiences propagate through memory. Conversely, individually noisy agents can generate stable population-level behavior through aggregation and network averaging. A small number of anomalous actions can initiate behavioral cascades when agents update their policies rapidly. Long memories may improve reciprocity but may also create persistent retaliation. Increased stochasticity may destabilize a population, but it may also prevent premature convergence to an undesirable equilibrium.

The scientific problem is therefore not simply:

Do LLM agents cooperate?

A more fundamental question is:

Under what combinations of memory, stochasticity, adaptation, population size, interaction frequency, game structure, and network topology does a population of LLM agents converge, oscillate, polarize, or become unstable?

This paper develops a framework for answering this question.

We define the *stability frontier* as the boundary in parameter space separating qualitatively different collective dynamical regimes. The frontier is not assumed to exist a priori. Rather, it is an empirical object to be estimated from repeated simulation.

The proposed framework is motivated by three bodies of literature.

First, behavioral game-theoretic research demonstrates that LLM agents do not necessarily behave as idealized rational players. Akata et al. [1] evaluated several LLMs in repeated 2×2 games and found systematic behavioral patterns in cooperation and coordination, demonstrating the value of game-theoretic analysis for LLM behavior.

Second, generative-agent research has shown that LLMs can produce persistent, memory-dependent, and socially interacting agents [9].

Third, classical complex-systems and evolutionary-game research has established that repeated interaction, network topology, adaptation, and stochasticity can generate emergent collective phenomena [2, 5, 8, 13].

The missing link is a unified framework connecting these microscopic mechanisms to macroscopic stability in LLM populations.

This paper makes five contributions:

1. It formulates interacting LLM agents as stochastic, history-dependent dynamical systems.
2. It defines measurable population-level quantities for cooperation, stability, adaptation, diversity, volatility, and resilience.
3. It proposes a multidimensional experimental design for identifying dynamical regime boundaries.
4. It introduces a probabilistic formulation of the stability frontier suitable for stochastic LLM systems.
5. It establishes a research protocol that can distinguish stable convergence from oscillation, polarization, stochastic drift, and genuine instability.

2 Related Work

2.1 LLMs as Strategic Agents

Classical game theory models decision makers as utility-maximizing agents selecting actions according to best responses or equilibrium concepts [7].

Repeated games extend this framework by allowing agents to condition future actions on previous outcomes, enabling phenomena such as reciprocity, punishment, forgiveness, and reputation [2].

LLM agents introduce a different regime. Their policies are generated through language-model inference and may depend on natural-language framing, contextual history, inferred intentions, and latent behavioral priors. They are therefore neither purely deterministic nor necessarily well described by classical expected-utility maximization.

Akata et al. [1] provide direct evidence for this distinction. Their experiments evaluated GPT-4, Claude 2, Llama 2 70B, and other models in repeated games. The study found substantial behavioral structure across game families, including cooperation and coordination patterns that differed across models and games.

This motivates treating an LLM policy as an empirical stochastic mapping rather than assuming an analytically known best-response function.

2.2 Generative Agents

Park et al. [9] introduced an architecture in which LLM-based agents maintain memories of observations, synthesize higher-level reflections, and use those memories to plan future actions. The resulting agents displayed persistent social behavior and emergent coordination in a simulated town.

This architecture is especially relevant to the present framework because memory is not simply a storage mechanism. It changes the effective state space of the agent.

If an agent’s action depends on its complete interaction history,

$$a_i(t) = f_i(h_i(t)),$$

then the process is generally non-Markovian with respect to the external environment alone.

2.3 Large-Scale Artificial Societies

Piao et al. [10] developed AgentSociety, a large-scale simulation platform for LLM-driven social agents. Their experiments involved more than 10^4 agents and approximately five million interactions, illustrating that synthetic societies can now be studied at substantially larger scales than early generative-agent systems.

This scaling changes the scientific regime. At large N , individual interactions may generate population-level feedback loops, cascades, polarization, and network effects.

2.4 Cooperation and Social Dilemmas

Cooperation has long been studied using repeated games, evolutionary dynamics, and network models [2, 8, 11]. In networked populations, cooperation can depend strongly on interaction structure, clustering, and strategy updating.

Recent surveys of multi-agent cooperation emphasize the importance of social dilemmas in understanding artificial and human-agent systems [6].

The present work extends these ideas to LLM agents whose strategies are generated by language models rather than fixed symbolic policies.

2.5 Complex Systems and Stability

Classical dynamical-systems research distinguishes fixed points, periodic orbits, chaotic trajectories, and stochastic processes [12]. In complex systems, increasing system size or interaction strength can lead to qualitative changes in collective dynamics [5].

The proposed stability frontier draws on this perspective while avoiding the assumption that LLM societies necessarily exhibit formal deterministic chaos. Because language-model sampling introduces stochasticity, the primary empirical concepts are *volatility*, *resilience*, *regime transition*, and *probabilistic instability*.

3 System Formulation

Consider a population of N interacting LLM agents,

$$\mathcal{A} = \{1, 2, \dots, N\}.$$

Time is discrete:

$$t = 0, 1, 2, \dots, \mathcal{T}.$$

Each agent i possesses:

- an observable environment state $s_i(t)$;
- an internal behavioral state $q_i(t)$;
- a finite interaction history $h_i^{(M)}(t)$;
- an action $a_i(t)$;
- a reward or payoff $r_i(t)$.

The interaction topology is represented by a graph

$$\mathcal{G} = (V, E),$$

where

$$|V| = N.$$

An edge $(i, j) \in E$ indicates that agents i and j can interact.
The environment also defines a game or payoff structure $\mathcal{H}$.
The complete system can therefore be represented by

$$\mathcal{S} = (N, M, T, F, \alpha, \mathcal{G}, \mathcal{H}).$$

4 Primary System Parameters

4.1 Population Size N

The population size is

$$N \in \mathbb{N}^+.$$

It controls the number of interacting decision makers.

Small populations allow detailed analysis of individual interactions, while large populations can exhibit collective phenomena that are difficult to observe in pairwise experiments.

We therefore consider

$$N \in \{2, 10, 50, 100, 500, 1000\},$$

with larger values possible depending on computational resources.

4.2 Memory Horizon M

The memory horizon determines how many previous interactions are available to the agent.

A simple history representation is

$$h_i^{(M)}(t) = \{(s_i(k), a_i(k), a_{-i}(k), r_i(k))\}_{k=\max(0, t-M)}^{t-1}.$$

For

$$M = 0,$$

the agent receives no explicit interaction history.

For larger M , historical information becomes available.

Importantly, memory capacity should not be confused with memory utilization. Two agents with the same context window may assign different effective importance to historical observations.

4.3 Stochasticity and Temperature T

LLM generation can be modeled approximately through a temperature-scaled softmax:

$$P(w_k \mid w_{<k}) = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)}.$$

As $T \rightarrow 0$, generation becomes increasingly concentrated on high-probability tokens, while larger values generally increase sampling variability.

However, temperature is only one source of behavioral stochasticity.

We therefore define an effective stochasticity parameter

$$\sigma_{\text{eff}} = f(T, \xi, \rho, \eta),$$

where:

- T is decoding temperature;
- ξ represents random sampling;
- ρ represents prompt sensitivity;
- η represents residual behavioral variation.

The empirical experiments should report both the configured temperature and observed action-level stochasticity.

4.4 Interaction Frequency F

Let

$$F > 0$$

denote the expected number of interactions per agent per macro-time step.

Increasing F increases the number of opportunities for information propagation and adaptation.

A useful interpretation is that F controls effective feedback strength:

$$\beta_{\text{feedback}} \propto F.$$

This relationship is not assumed to be linear and should be measured experimentally.

4.5 Adaptation Rate α

The adaptation coefficient is

$$\alpha \in [0, 1].$$

A generic update rule is

$$q_i(t+1) = (1 - \alpha)q_i(t) + \alpha\hat{q}_i(t),$$

where $\hat{q}_i(t)$ is the strategy estimate produced by the agent's reflection or learning process.

Low α corresponds to slow behavioral updating.

High α corresponds to rapid incorporation of recent evidence.

This parameter may generate a classical feedback trade-off:

$$\text{slow adaptation} \longleftrightarrow \text{robustness} \longleftrightarrow \text{over-correction}.$$

5 Stochastic Agent Policy

The action of agent i is generated according to

$$a_i(t) \sim \pi_i \left(a \mid s_i(t), h_i^{(M)}(t), q_i(t), T \right).$$

The policy π_i is not assumed to have a closed-form analytical expression. For binary cooperation games,

$$A = \{C, D\},$$

we define

$$x_i(t) = P(a_i(t) = C).$$

The population strategy state is

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_N(t) \end{bmatrix}.$$

The population evolves according to

$$\mathbf{x}(t+1) = \mathcal{F} \left(\mathbf{x}(t), H^{(M)}(t), T, F, \alpha, \mathcal{G}, \mathcal{H}, \xi_t \right).$$

Here ξ_t represents stochastic variation.

The central methodological challenge is that $\mathcal{F}$ is induced by an LLM and therefore generally cannot be written analytically.

The objective of the experiment is consequently to estimate properties of $\mathcal{F}$ from repeated trajectories.

6 Macroscopic Observables

6.1 Cooperation Rate

For binary actions,

$$C(t) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[a_i(t) = C].$$

The long-run cooperation level is

$$\overline{C} = \frac{1}{\mathcal{T} - t_0} \sum_{t=t_0}^{\mathcal{T}} C(t),$$

where t_0 is the end of the transient period.

6.2 Behavioral Stability

Define population behavioral displacement as

$$V(t) = \frac{1}{N} \sum_{i=1}^N |x_i(t) - x_i(t-1)|.$$

A normalized behavioral stability score is

$$S(t) = 1 - V(t).$$

Thus,

$$S(t) \rightarrow 1$$

indicates small short-term policy changes.

However, $S(t) \approx 1$ does not necessarily imply mathematical equilibrium. A population can be stable while remaining far from a game-theoretic equilibrium.

For this reason, we distinguish:

1. behavioral stability;
2. equilibrium proximity;
3. perturbation resilience.

6.3 Adaptation Velocity

Following a perturbation at time t_0 ,

$$A_{\text{vel}} = \frac{1}{N\Delta t} \sum_{i=1}^N |x_i(t_0 + \Delta t) - x_i(t_0)|.$$

Large A_{vel} indicates rapid response.

However, rapid response is not automatically desirable. A high adaptation velocity may correspond either to effective adaptation or to unstable overreaction.

6.4 Recovery Time

Define recovery time as

$$\tau_{\text{rec}} = \inf \{ \tau : d(t_0 + \tau) < \epsilon \},$$

where

$$d(t) = \frac{1}{N} \sum_i |x_i(t) - x_i^{\text{baseline}}(t)|.$$

This provides a direct measure of resilience.

6.5 Strategic Diversity

Suppose agents are assigned to K behavioral strategy clusters.

Let

$$p_k(t)$$

be the proportion of agents in cluster k .

Shannon diversity is

$$D(t) = - \sum_{k=1}^K p_k(t) \log_2 p_k(t).$$

Normalized diversity is

$$D_{\text{norm}} = \frac{D}{\log_2 K}.$$

High diversity represents behavioral heterogeneity rather than instability per se.

6.6 Behavioral Volatility

We define

$$V_C = \text{Var}[C(t)].$$

High volatility indicates large fluctuations in aggregate behavior.

6.7 Autocorrelation

Behavioral persistence can be measured using

$$R_C(k) = \text{Corr}[C(t), C(t - k)].$$

Long-lived autocorrelation indicates persistent population-level behavior.

7 Polarization and Network-Level Heterogeneity

Global cooperation can conceal important internal structure.

For example,

$$C(t) = 0.5$$

could correspond either to every agent cooperating with probability 0.5 or to two stable communities in which one group cooperates and the other defects.

We therefore introduce a polarization measure.

Let

$$x_i(t)$$

be the cooperation propensity of agent i .

Define

$$P(t) = \text{Var}_i[x_i(t)].$$

Network polarization can additionally be defined through edge-level disagreement:

$$P_{\mathcal{G}} = \frac{1}{|E|} \sum_{(i,j) \in E} |x_i - x_j|.$$

This distinguishes:

- global instability;
- stable heterogeneity;
- community polarization.

8 Dynamical Regimes

We classify population trajectories into five primary regimes.

8.1 Stable Regime

A population is classified as stable if

$$\bar{S} > \theta_S,$$

$$V_C < \theta_V,$$

and perturbation recovery occurs within the observation horizon.

8.2 Adaptive Regime

A population is adaptive if it responds significantly to perturbations but returns toward its baseline state.

8.3 Oscillatory Regime

A population is oscillatory if

$$C(t)$$

contains persistent periodic or quasi-periodic components.

Spectral density can be used to detect dominant frequencies.

8.4 Polarized Regime

A population is polarized if

$$P(t) > \theta_P$$

while the aggregate cooperation rate remains comparatively stable.

8.5 Unstable Regime

A population is classified as dynamically unstable if small perturbations produce persistent macroscopic deviations and no convergence is observed within the experimental horizon.

Importantly, instability should not be equated automatically with deterministic chaos.

9 The Stability Frontier

The central object is

$$\Phi = \Phi(M, T, N, \alpha, F, \mathcal{G}, \mathcal{H}).$$

The frontier is defined implicitly by

$$\Phi = 0.$$

However, because LLM systems are stochastic, a deterministic boundary may be inappropriate.

We therefore define the probability of entering an unstable regime:

$$P_{\text{unstable}} = f(M, T, N, \alpha, F, \mathcal{G}, \mathcal{H}).$$

The empirical frontier can then be defined by

$$P_{\text{unstable}} = p_c,$$

where a natural reference value is

$$p_c = 0.5.$$

This converts the phase boundary into a statistical classification problem. The resulting phase surface is

$$\Phi(M, T, N, \alpha, F, \mathcal{G}, \mathcal{H}) = P_{\text{unstable}} - p_c$$

with

$$\Phi < 0$$

corresponding to a predominantly stable region and

$$\Phi > 0$$

corresponding to a predominantly unstable region.

10 Research Hypotheses

10.1 H1: Memory

Increasing memory will initially increase behavioral persistence and may increase cooperation in repeated social dilemmas.

However, sufficiently strong memory may produce path dependence and persistent retaliation.

The expected relationship is therefore potentially non-monotonic:

$$M \uparrow \Rightarrow S \uparrow \quad \text{initially,}$$

followed by

$$S \downarrow$$

when memory-induced feedback becomes excessive.

10.2 H2: Stochasticity

Increasing temperature is expected to increase action-level variance.

Moderate stochasticity may increase strategic exploration and prevent premature convergence, while sufficiently high stochasticity may reduce temporal coherence.

We therefore hypothesize a regime boundary

$$T_c = T_c(M, N, \alpha, F, \mathcal{G}, \mathcal{H})$$

rather than a universal critical temperature.

10.3 H3: Population Size

Increasing N may either stabilize aggregate behavior through averaging or destabilize the system through additional feedback pathways.

We therefore test

$$\frac{\partial S}{\partial N}$$

empirically rather than assuming its sign.

10.4 H4: Adaptation

We hypothesize an inverted-U relationship between adaptation rate and stability:

$$S(\alpha)$$

may be maximal at an intermediate value

$$\alpha^*.$$

Very low α may produce sluggish adaptation, while very high α may produce over-correction.

10.5 H5: Interaction Frequency

Increasing F is expected to strengthen feedback.

Therefore,

$$F \uparrow$$

may amplify both stabilizing and destabilizing mechanisms depending on the underlying game and topology.

10.6 H6: Network Topology

Network topology will modify the stability frontier.

Dense networks may increase information propagation, while modular networks may isolate disturbances and permit stable coexistence of distinct behavioral communities.

11 Experimental Design

11.1 Experiment 1: Memory Sweep

$$M \in \{0, 1, 5, 10, 20, 50\}.$$

Baseline:

$$N = 100, \quad T = 0.2, \quad \alpha = 0.3, \quad F = 1.$$

Measurements:

$$\overline{C}, \quad \overline{S}, \quad D, \quad P, \quad \tau_{\text{rec}}.$$

11.2 Experiment 2: Stochasticity Sweep

$$T \in \{0, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}.$$

Baseline:

$$N = 100, \quad M = 5, \quad \alpha = 0.3, \quad F = 1.$$

Measurements:

$$\overline{S}, \quad V_C, \quad D, \quad R_C.$$

11.3 Experiment 3: Population Scale

$$N \in \{2, 10, 50, 100, 500, 1000\}.$$

Baseline:

$$M = 5, \quad T = 0.2, \quad \alpha = 0.3, \quad F = 1.$$

Measurements:

$$D, \quad S, \quad P, \quad V_C.$$

11.4 Experiment 4: Adaptation

$$\alpha \in \{0.05, 0.1, 0.3, 0.5, 0.7, 0.9\}.$$

Measurements:

$$A_{\text{vel}}, \quad S, \quad \tau_{\text{rec}}, \quad V_C.$$

11.5 Experiment 5: Interaction Frequency

$$F \in \{0.25, 0.5, 1, 2, 5\}.$$

This experiment tests whether increased interaction frequency amplifies feedback.

11.6 Experiment 6: Network Topology

The following graphs should be compared:

1. complete graph;
2. Erdős–Rényi random graph;
3. Watts–Strogatz small-world network;
4. Barabási–Albert scale-free network;
5. modular community network;
6. spatial lattice.

12 Game Environments

The framework should be evaluated across multiple game classes.

12.1 Iterated Prisoner's Dilemma

The Prisoner's Dilemma provides the primary cooperation benchmark.

A canonical payoff matrix is

	C	D
C	R	S
D	T	P

with

$$T > R > P > S.$$

Repeated interaction allows reciprocal strategies to emerge.

12.2 Stag Hunt

The Stag Hunt tests coordination under multiple equilibria.

It is useful for distinguishing cooperation from coordination.

12.3 Battle of the Sexes

Battle of the Sexes introduces conflicting preferences over coordinated outcomes.

It is particularly relevant to measuring population-level coordination failure.

12.4 Public Goods Game

The Public Goods Game introduces a many-agent social dilemma in which individual contributions generate collective benefits.

This provides a direct test of whether pairwise cooperation mechanisms generalize to larger populations.

13 Perturbation and Resilience Experiments

Stability should not be measured only through passive observation.

We therefore introduce controlled perturbations.

At time

$$t = t_0,$$

one of the following shocks is applied:

1. forced defection by a fraction of agents;
2. sudden payoff modification;

3. removal of a network community;
4. introduction of a high-temperature interval;
5. sudden increase in adaptation rate;
6. temporary memory truncation.

The response is measured using

$$d(t) = \frac{1}{N} \sum_i |x_i(t) - x_i^{\text{baseline}}(t)|.$$

Important quantities include

$$d_{\max} = \max_t d(t)$$

and

$$\tau_{\text{rec}}.$$

A resilient population should exhibit bounded displacement and finite recovery.

14 Statistical Identification

Each experimental configuration should be replicated across independent random seeds.

Let

$$r = 1, \dots, R$$

denote replicate simulations.

For outcome Y , we propose the hierarchical model

$$Y_r = \beta_0 + \beta_M M + \beta_T T + \beta_N \log(N) + \beta_\alpha \alpha + \beta_F F + u_{\text{model}} + u_{\text{seed}} + \epsilon_r.$$

Interaction terms should include

$$M \times T,$$

$$M \times \alpha,$$

$$T \times \alpha,$$

$$N \times F,$$

and

$$\alpha \times F.$$

The log transformation of N is useful because population effects may exhibit diminishing returns.

For instability classification, we propose logistic regression:

$$\text{logit}(P_{\text{unstable}}) = \beta_0 + \sum_j \beta_j X_j + \sum_{j < k} \beta_{jk} X_j X_k.$$

This model directly estimates the probability of entering an unstable regime.

15 Detecting Phase Transitions

A visually sharp change in a metric does not constitute sufficient evidence for a phase transition.

A candidate transition should satisfy several criteria.

1. The transition should be reproducible across random seeds.
2. The transition should remain visible under increased simulation duration.
3. Neighboring parameter values should produce statistically distinct regimes.
4. The transition should persist across independent simulation batches.
5. Multiple observables should change consistently.

Potential early-warning signals include:

$$\text{Var}[C(t)] \uparrow$$

and

$$R_C(k) \uparrow$$

near a critical regime.

Such increases may indicate critical slowing down or increasing persistence of fluctuations, concepts commonly studied in complex dynamical systems.

16 Experimental Pipeline

A complete experiment consists of six stages.

1. Initialize the population and network.
2. Assign system parameters.
3. Execute repeated game interactions.
4. Update memory and adaptive states.
5. Log individual and population trajectories.
6. Compute dynamical observables and classify the regime.

The following pseudocode summarizes the process.

Algorithm 1 LLM Society Stability Experiment

Require: $N, M, T, F, \alpha, \mathcal{G}, \mathcal{H}, \mathcal{T}, R$

```

1: for  $r = 1$  to  $R$  do
2:   Initialize agents  $\mathcal{A}$ 
3:   Initialize network  $\mathcal{G}$ 
4:   Initialize histories  $h_i^{(M)}$ 
5:   for  $t = 1$  to  $\mathcal{T}$  do
6:     Select interacting pairs according to  $\mathcal{G}$ 
7:     for each interaction  $(i, j)$  do
8:       Construct agent prompts
9:       Generate actions  $a_i(t), a_j(t)$ 
10:      Compute payoffs  $r_i(t), r_j(t)$ 
11:      Update histories
12:    end for
13:    Apply adaptation update with  $\alpha$ 
14:    Log population state
15:  end for
16:  Compute  $C, S, A_{\text{vel}}, D$ 
17:  Compute volatility, polarization, and recovery
18:  Classify dynamical regime
19: end for

```

Ensure: Empirical stability surface

17 Reproducibility Protocol

Every simulation should record:

- model identifier and version;

- system prompt;
- game specification;
- payoff matrix;
- decoding parameters;
- random seed;
- memory contents;
- network topology;
- interaction schedule;
- agent actions;
- rewards;
- adaptation events;
- token budget;
- simulation timestamp;
- software version.

The complete trajectory should be stored rather than only aggregate statistics.

This is essential because post-hoc strategy classification can reveal mechanisms that are invisible in population-level averages.

18 Ablation Studies

To establish causal mechanisms, each component should be systematically removed or controlled.

18.1 Memory Ablation

Compare:

$$M = 0$$

against equivalent agents receiving the same prompt without historical information.

18.2 Reflection Ablation

Remove explicit reflection or strategy-update steps while keeping interaction history unchanged.

18.3 Adaptation Ablation

Set

$$\alpha = 0.$$

This produces a non-adaptive population.

18.4 Stochasticity Ablation

Compare deterministic decoding with stochastic sampling.

18.5 Network Ablation

Replace structured networks with randomized graphs preserving approximately the same degree distribution.

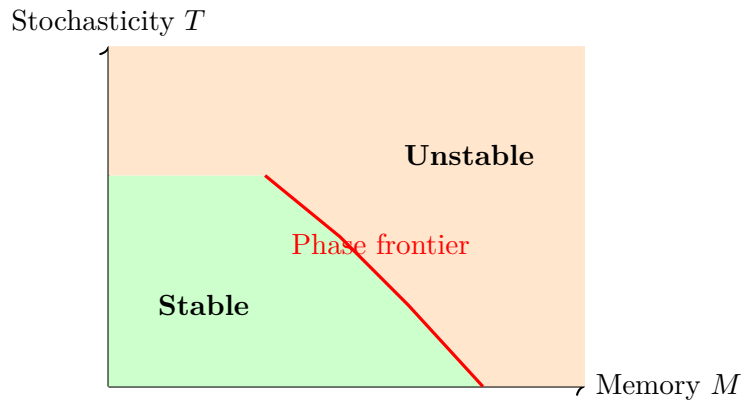
These experiments distinguish the effect of the nominal parameter from confounding implementation details.

19 Expected Phase Structure

The initial two-dimensional phase diagram can be constructed in the (M, T) plane while fixing

$$N, \alpha, F, \mathcal{G}, \mathcal{H}.$$

Conceptually:



The figure is conceptual rather than a prediction.
The actual frontier must be estimated from simulation.

20 Finite-Size Effects and Scaling

A critical question is whether observed transitions persist as population size increases.

Let

$$N_1 < N_2 < \dots < N_k.$$

For each population size, estimate the transition location

$$T_c(N).$$

If

$$\lim_{N \rightarrow \infty} T_c(N)$$

exists, the system may exhibit a meaningful large-population stability threshold.

If instead

$$T_c(N)$$

continues drifting with N , the observed transition may be a finite-size phenomenon.

This distinction is essential for claims about artificial societies at deployment scale.

21 Universality and Model Dependence

A central scientific question is whether the stability frontier is universal across LLM families.

Let

$$\Phi_m$$

denote the frontier for model family m .

A universal frontier would approximately satisfy

$$\Phi_m \approx \Phi_{m'}$$

after suitable normalization.

Alternatively, different models may produce distinct frontiers:

$$\Phi_m \neq \Phi_{m'}.$$

Model dependence should therefore be treated as a first-class scientific result rather than noise.

Possible sources include:

- pretraining distribution;
- instruction tuning;
- alignment procedures;
- context-window behavior;
- reasoning style;
- decoding characteristics;
- implicit social priors.

22 Limitations

Several limitations constrain interpretation.

First, temperature is not equivalent to behavioral stochasticity. Different decoding procedures can produce different action distributions at the same temperature.

Second, memory length M does not capture the semantics of memory retrieval. Two agents with identical memory windows may use historical events very differently.

Third, the adaptation parameter α is a modeling abstraction. Actual LLM adaptation may occur through complex natural-language reasoning rather than a simple linear update.

Fourth, an LLM’s policy may change because of prompt structure even when the nominal system variables are held constant.

Fifth, finite simulation horizons can produce false impressions of convergence. A system that appears stable over 100 interactions may become unstable over 10^4 interactions.

Sixth, the definition of “chaos” requires particular care. Stochastic volatility should not be confused with deterministic chaos.

Finally, synthetic societies are not automatically valid models of human societies. The framework is intended primarily as a scientific methodology for studying *LLM-based artificial societies*, not as a direct replacement for empirical human social science.

23 Safety and Governance Implications

The proposed framework has potential relevance to the deployment of autonomous multi-agent systems.

If particular combinations of memory, stochasticity, interaction frequency, and adaptation are consistently associated with instability, deployment systems could be constrained to remain inside empirically validated safety envelopes.

For example, a system could require

$$P_{\text{unstable}} < \delta,$$

where δ is an application-specific risk threshold.

Such a constraint would not guarantee safety. However, it would provide a quantitative method for measuring dynamical risk rather than relying solely on individual-agent benchmarks.

The framework may therefore support a transition from qualitative multi-agent safety testing toward measurable population-level resilience testing.

24 Discussion

The central proposition of this work is that LLM societies should be studied as dynamical systems rather than as collections of independent language models.

At the microscopic level, an agent may appear relatively stable.

At the macroscopic level, however, the population may exhibit:

feedback $\rightarrow$ adaptation $\rightarrow$ behavioral change $\rightarrow$ new observations $\rightarrow$ feedback.

This feedback loop is the mechanism through which small individual-level differences can become large population-level effects.

Memory determines how long information remains behaviorally relevant.

Stochasticity determines how much variation enters the system.

Adaptation determines how rapidly the system responds.

Interaction frequency determines how rapidly feedback propagates.

Population size determines the scale at which local events can become collective phenomena.

Network topology determines which agents influence one another.

Game structure determines the incentives governing those interactions.

The resulting system is therefore fundamentally multidimensional.

A major implication is that there may be no single scalar “temperature threshold” or “memory threshold.” Instead, the critical condition may depend on combinations such as

$$T_c = T_c(M, \alpha, N, F, \mathcal{G}, \mathcal{H}).$$

This suggests that the stability frontier should be understood as a high-dimensional surface rather than a single number.

25 Conclusion

This paper introduced a mathematical and computational framework for studying the stability of LLM-based artificial societies.

The framework models LLM agents as stochastic, history-dependent decision processes and identifies five principal system variables:

$$(N, M, T, F, \alpha).$$

These variables determine how agents remember, act, adapt, and interact. Four primary macroscopic observables are proposed:

$$C, \quad S, \quad A_{\text{vel}}, \quad D.$$

They are supplemented by volatility, polarization, autocorrelation, and perturbation recovery.

The central research object is the empirical stability surface

$$\boxed{\Phi(M, T, N, \alpha, F, \mathcal{G}, \mathcal{H}) = 0}$$

which separates qualitatively different population regimes.

Importantly, the existence of such a surface is treated as an empirical hypothesis.

The proposed experimental methodology allows researchers to test whether LLM populations exhibit stable, adaptive, oscillatory, polarized, or unstable regimes; whether these regimes persist as populations scale; and whether the resulting phase structure generalizes across models, games, and network topologies.

The broader scientific objective is to establish a systems-level methodology for studying collective behavior in artificial societies.

As LLM agents become increasingly autonomous and increasingly interconnected, understanding the behavior of individual agents will remain necessary but will no longer be sufficient.

The relevant scientific unit will increasingly become the *population*.

References

- [1] Akata, E., Schulz, L., Coda-Forno, J., Oh, S. J., Bethge, M., and Schulz, E. Playing repeated games with large language models. *Nature Human Behaviour*, 9:1380–1390, 2025. doi:10.1038/s41562-025-02172-y.
- [2] Axelrod, R. and Hamilton, W. D. The evolution of cooperation. *Science*, 211(4489):1390–1396, 1981. doi:10.1126/science.7466396.
- [3] Barabási, A.-L. and Albert, R. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. doi:10.1126/science.286.5439.509.

- [4] Griffiths, T. L., Callaway, F., Chang, M. B., Grant, E., Krueger, P. M., and Lieder, F. Doing more with less: Meta-reasoning and meta-learning. *Behavioral and Brain Sciences*, 2019.
- [5] May, R. M. Simple mathematical models with very complicated dynamics. *Nature*, 261:459–467, 1976. doi:10.1038/261459a0.
- [6] Mu, C., Guo, H., Chen, Y., Shen, C., Hu, S., and Wang, Z. Multi-agent, human-agent and beyond: A survey on cooperation in social dilemmas. *Neurocomputing*, 610:128514, 2024. doi:10.1016/j.neucom.2024.128514.
- [7] Nash, J. Equilibrium points in n -person games. *Proceedings of the National Academy of Sciences*, 36(1):48–49, 1950. doi:10.1073/pnas.36.1.48.
- [8] Nowak, M. A. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006. doi:10.1126/science.1133755.
- [9] Park, J. S., O’Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. doi:10.1145/3586183.3606763.
- [10] Piao, J., Yan, Y., Zhang, J., Li, N., Yan, J., Lan, X., Lu, Z., Zheng, Z., Wang, J. Y., Zhou, D., Gao, C., Xu, F., Zhang, F., Rong, K., Su, J., and Li, Y. AgentSociety: Large-scale simulation of LLM-driven generative agents advances understanding of human behaviors and society. arXiv preprint arXiv:2502.08691, 2025.
- [11] Szabó, G. and Fáth, G. Evolutionary games on graphs. *Physics Reports*, 446(4–6):97–216, 2007. doi:10.1016/j.physrep.2007.04.004.
- [12] Strogatz, S. H. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. CRC Press, 2nd edition, 2018.
- [13] Watts, D. J. and Strogatz, S. H. Collective dynamics of small-world networks. *Nature*, 393:440–442, 1998. doi:10.1038/30918.
- [14] Wooldridge, M. and Jennings, N. R. Intelligent agents: Theory and practice. *The Knowledge Engineering Review*, 10(2):115–152, 1995. doi:10.1017/S0269888900008122.